Md Arshad Ali
Data Scientist

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Professional Summary

Experienced Data Scientist with a strong background in Machine Learning, Deep Learning, NLP, and Generative AI. Skilled in building predictive models, automation pipelines, and RAG-based GenAI solutions. Hands-on expertise in LLM fine-tuning, prompt engineering, vector databases, and tools like LangChain, LlamaIndex, and OpenAI APIs. Proven ability to deliver production-ready, data-driven solutions that drive business impact and innovation.

Skills

Technical Skills

- **Machine Learning:** Regression, Classification, KNN, Predictive Modeling, Ensemble Methods, Model Deployment, MLOps
- **Deep Learning:** Neural Networks, CNNs, RNNs, LSTM, NLP, Generative AI, GANs, Transformers
- **Generative AI & LLMs:** RAG (Retrieval-Augmented Generation), LangChain, LLaMA, OpenAI GPT, Prompt Engineering, Fine-tuning, Vector Databases, Hugging Face
- **Vector Databases**: FAISS, ChromaDB, Weaviate
- **MLOps & Deployment**: MLflow, Docker, Kubernetes, FastAPI, Streamlit, Model Monitoring, GitHub Actions
- **Programming & Frameworks:** Python, R, SQL, PyTorch, TensorFlow, Keras, VBA
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau, Power BI
- **Data Engineering:** ETL Pipelines, MongoDB, PostgreSQL, NoSQL
- **Cloud Platforms**: AWS, GCP, Azure
- **Tools & Platforms:** Scikit-learn, Alteryx, Advanced Excel, Git, Docker, Jupyter, VS Code, Postman

Soft Skills

- Team Collaboration, Leadership, Strategic Decision-Making

Senior Technical Lead

Coforge **06/2025 – Present**

- Led development of an end-to-end **AI-enabled SDLC platform**, collaborating closely with AI experts and cross-functional teams from ideation to production.
- Designed and implemented multiple **Model Context Protocol (MCP)** tools to standardize AI interactions across different stages of the software development lifecycle.
- Built and deployed several **Uvicorn-based backend services** to support frontend applications, ensuring scalable, low-latency API communication.
- Contributed to both **frontend and backend development**, enabling seamless integration between user interfaces, APIs, and AI services.
- Integrated **Generative AI capabilities** across the SDLC, including requirements analysis, code assistance, validation workflows, and intelligent automation.
- **Architected modular**, production-ready services to support real-time AI inference, tool execution, and orchestration.
- Acted as **technical lead**, guiding design decisions, reviewing code, and ensuring best practices in scalability, security, and maintainability.
- Enabled faster development cycles and improved engineering productivity by **embedding AI-driven workflows** into the platform.
- Developed an **AI-powered chatbot with real-time database** integration for direct data retrieval, enabling dynamic, context-aware responses.
- Built a **RAG (Retrieval-Augmented Generation)** based chatbot to enhance information retrieval accuracy and reduce hallucinations in client-facing use cases.

Data Scientist
Leewayhertz Technologies Pvt. Ltd. **04/2025 – 06/2025**

- Contributed to internal **GenAI POCs** and research discussions around **Agentic AI and LangChain** agents.
- Assisted in refining prompt templates and **evaluating LLM output quality** across different use cases.
- Helping team on a **churn prediction project**, focusing on identifying key drivers of customer attrition using statistical modeling techniques.

Assistant Manager(Data Scientist)
Genpact India Pvt. Ltd. **01/2022 – 04/2025**

- **Sales Forecasting & Risk Modeling**: Built scalable machine learning models (XGBoost, LightGBM, Prophet) for sales prediction, customer churn, and risk scoring, improving forecast accuracy and reducing decision-making time by 35%.
- **End-to-End ML Pipelines**: Designed and automated full ML pipelines using Airflow, Docker, and MLflow, reducing manual deployment overhead by 40%.
- **Time Series Deployment:** Deployed time series models to GCP Vertex AI, enabling real-time prediction APIs for dynamic decision-making.
- **RAG-based Document Assistant:** Developed a document Q&A and retrieval system using LangChain, FAISS, and OpenAI GPT-3.5, enhancing document search efficiency for underwriters by 60%.
- **LangChain Agents:** Automated document summarization, classification, and extraction tasks with LangChain agents, reducing manual review time by 50%.
- **Agentic AI Prototype:** Designed a dynamic Agentic workflow using OpenAI tools (Function Calling, Memory, Tool Use) for an insurance customer support bot capable of adaptive conversation and automated claim handling.

Data Scientist – AXA XL **10/2021 – 01/2022**

- **Underwriting Accuracy:** Designed ML models to improve underwriting accuracy by 30% and reduce claims by 20%.
- **Fraud Prevention:** Developed fraud detection algorithms, lowering fraudulent claims by 25%.
- **Project Leadership:** Managed 5+ concurrent projects, achieving 100% on-time delivery.

Senior Analyst(Data Scientist) – Genpact India Pvt. Ltd. **02/2021 – 10/2021**

- **Data Profiling & Analysis**: Performed deep exploratory data analysis (EDA) using Pandas, NumPy, and Seaborn, discovering inefficiencies and trends in operational data.
- **Strategic Insights:** Built dashboards and visual stories with Power BI and Tableau, helping clients optimize operations and improve decision turnaround by 20%.
- **ETL Automation:** Automated data pipelines using Python, SQL and VBA reducing manual work by 50% and cutting data issues by 40%.

Data Analyst – Deloitte **08/2018 – 02/2021**

- **Data-Driven Insights:** Conducted ad hoc and monthly analytics using SQL, Excel, and Power BI for regulatory reporting and risk assessments.
- **Operational Reporting:** Developed KPI dashboards for business units, improving performance tracking and reporting speed by 30%.
- **Process Automation:** Reduced reporting cycle times by automating recurring reports and ETL processes using Python scripting and VBA Macros.

EDUCATION

Indira Gandhi National Open University (IGNOU) – India 2022 –2024
Master of Business Administration (MBA)

University of Delhi – India 2014 –2017
Bachelor of Commerce (B.Com)

Personal Projects

Movie Recommendation System

- Built and fine-tuned a movie recommender system to deliver highly accurate and relevant suggestions.

- Evaluated **multiple algorithms, achieving optimal performance and enhancing user engagement.**

E-Commerce Exploratory Data Analysis
- Conducted in-depth EDA on e-commerce datasets to identify trends and inform strategic decisions.
- Provided actionable insights for customer segmentation, pricing strategies, and marketing campaigns.

Achievements

- Improved underwriting processes with ML models, reducing claim incidents by 20%.
- Automated 50% of manual tasks, improving efficiency and reducing errors across multiple roles.
- Successfully managed and delivered 5+ high-impact projects on schedule in a fast-paced environment.